# Can Large Language Models Recognize Social Anxiety in Text? A Comparative Classification and Bias-Probing Study for Teen-Facing AI

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***Abstract*—Large language models (LLMs) increasingly encounter text expressing psychological distress, yet little is known about whether they can distinguish *social anxiety* – fear of negative evaluation by others – from superficially similar states like general anxiety and sadness. We frame this as a text classification and robustness problem, not a diagnostic one. Using proxy labels built from EmpatheticDialogues and GoEmotions and grounded in the Social Anxiety Scale for Adolescents (SAS-A), we evaluate six LLMs across three vendors and two cost tiers on a balanced four-class task. All six recognize sadness reliably (F1 0.75–0.86) but show a persistent social-anxiety recall gap (0.38–0.60) that does not close with a more capable tier. A counterfactual bias probe (same content, varied age/gender descriptors and register) first hit a ceiling (100% accuracy, 0% prediction change); deliberately ambiguous templates broke it (60–87.5% accuracy, 5–20% demographic flip rate) and suggested, at small $n$, a register effect in two models. A role-framed few-shot prompt makes every model *worse*; real teen-authored blog posts are classified *more* accurately than adult-authored ones; and a DistilBERT classifier fine-tuned on 300 examples per class reaches 0.89 social-anxiety recall, so the signal is learnable from these labels. We discuss implications for bias-probe design, prompt selection, and teen-facing conversational AI.**

***Index Terms*—large language models, text classification, social anxiety, adolescent mental health, algorithmic bias, counterfactual evaluation**

## I. Introduction

Conversational AI systems used by teenagers – school chatbots, moderation pipelines, wellness apps – routinely process text that may carry signs of psychological distress. A large fraction of that distress in adolescence is socially mediated: fear of being judged by peers, avoidance of unfamiliar groups, rumination about how one is perceived [1]. Whether current large language models (LLMs) can tell *social anxiety* apart from general worry or sadness is therefore a practically relevant question, independent of whether LLMs should ever be used for anything resembling diagnosis – a use we explicitly do not endorse or evaluate.

This paper treats the question as one of **text classification capability and robustness**. We ask: (1) Can six current LLMs distinguish social anxiety from general anxiety, sadness, and neutral text? (2) Does a more capable tier from the same vendor close any gap? (3) Are predictions stable when the same content is attributed to speakers of different ages, genders, and registers?

Our contributions are: (a) a proxy-label construction grounded in a validated adolescent psychometric framework (SAS-A); (b) a six-model, two-tier comparative evaluation showing a persistent, tier-invariant recall gap specifically on social anxiety; (c) a difficulty-graded counterfactual probe design, showing that a *ceiling effect* on clean minimal pairs can mask a demographic bias signal and that adding ambiguous items surfaces one; and (d) three further checks – prompt sensitivity, real-age ecological validation, and a fine-tuned baseline – attributing the gap to zero-shot prompting rather than an unlearnable signal. The contribution is primarily an evaluation; we report its uncertainty explicitly and treat small-sample effects as leads, not conclusions.

## II. Related Work

**LLM mental-health text classification.** Prior work on computational detection of psychological states from text has largely used coarse categories (stress, depression, generic anxiety) on social-media corpora such as Dreaddit [2] and has not isolated social anxiety as a construct distinct from general anxiety or sadness. More recent LLM-specific studies evaluate anxiety/depression/stress detection and prompting strategy [8], within a growing survey literature [9]; none isolate social anxiety from general anxiety or report a role/few-shot prompt *degrading* performance as we find (Section V-C).

**Adolescent social anxiety.** La Greca and Lopez's Social Anxiety Scale for Adolescents (SAS-A) [1] organizes social anxiety into Fear of Negative Evaluation, Social Avoidance and Distress-General, and Social Avoidance specific to new situations – a three-factor structure we use to ground our synthetic scenario design (Section III) without reproducing any copyrighted scale item.

**Counterfactual bias probing.** Minimal-pair methods (e.g., WinoBias gender-swap probes [7]) isolate a demographic

variable's causal effect, which natural text confounds with content. We apply the same logic to age, gender, and register.

**Teen-facing conversational AI.** Recent work shows LLM chatbots used by minors often fail on content- and crisis-related safety: MinorBench finds general-purpose LLMs poorly refuse unsafe content aimed at children [11], and a multi-model study of mental-health crisis handling finds inconsistent, sometimes harmful responses [12]. Both probe chatbot *output* safety; we instead probe upstream *recognition* accuracy – whether a model can identify social anxiety at all, a prerequisite for any downstream safety behavior.

## III. Data and Construct Definition

### A. Proxy labels for a four-class task

No public dataset ships a `social_anxiety` label. We construct one by combining two datasets with complementary properties. From **EmpatheticDialogues** [3] – 25k conversations, each grounded in a first-person situation the speaker wrote to describe when they felt a given emotion – we route the `afraid / anxious / apprehensive / terrified / embarrassed` labels through a social-context keyword filter (presence of terms such as *class, party, everyone, group chat*) to split them into `social_anxiety` and `general_anxiety`, and route `sad / devastated / disappointed / lonely` to `sadness`. From **GoEmotions** [4] – 58k Reddit comments – we take the native `neutral` label, since EmpatheticDialogues has no neutral category.

Sourcing `social_anxiety` from GoEmotions as well failed: its comments are mostly reactions, not first-person disclosures, and the same filter yielded 46 examples from 58k comments (vs. 421 from EmpatheticDialogues). Because `neutral` is drawn from a different corpus than the other three classes, it differs from them in genre as well as emotional content – a limitation detailed in Section VII. This does not affect the three-way contrast among `social_anxiety`, `general_anxiety`, and `sadness`, which are homogeneously sourced. A two-rater audit (Section VII) finds `social_anxiety` the noisiest class, so all results measure agreement with a proxy construct, not clinical judgment. We sample a balanced 300-item evaluation set (75 per class).

### B. Counterfactual templates for the bias probe

We write $N = 24$ original scenario templates, six per class, each rendered in a formal register and a teen-digital register, and each instantiated with four demographic descriptors ("a 15-year-old girl," "a 15-year-old boy," "a 35-year-old woman," "a 35-year-old man"), for $24 \times 2 \times 4 = 192$ stimuli. Social-anxiety templates are distributed across the three SAS-A factors (Fear of Negative Evaluation, Social Avoidance-General, Social Avoidance-New) to ground the construct theoretically; no SAS-A item wording is reproduced.

For example, a Fear-of-Negative-Evaluation template reads (formal): *"A 15-year-old girl kept replaying the conversation from lunch, convinced her classmates thought the joke she made was stupid"*; its teen-digital rendering reads *"[Subject] literally could not stop replaying that lunch conversation, ngl pretty sure the whole class thought that joke was cringe."* All 8 renderings share one event.

Evaluating these 192 stimuli across all six models produced a **ceiling effect**: 100% classification accuracy and 0% prediction change across demographic variants, for every model (Section V-B), which cannot distinguish "no bias" from "too easy to reveal bias." We therefore add ten deliberately *ambiguous* templates straddling the social/general anxiety boundary (4), the social anxiety/sadness boundary (4), and general anxiety/sadness (2), e.g., *"A 15-year-old girl stayed up rehearsing the presentation, worried the group would think her slides weren't good enough"* – a scenario compatible with either fear of social evaluation or generic performance worry. Because these items are intentionally ambiguous, we treat the assigned label as the researcher's best-guess primary reading rather than a confident gold label, and rely on the *prediction-flip rate* – a label-free metric – as the primary bias signal: a scenario-register unit "flips" if any of its four demographic variants receives a different label from the others, and the flip rate is the fraction of the $10 \times 2 = 20$ units that flip.

### C. Real-age ecological complement

The templates in Section III-B isolate demographic effects causally but rely on an author *descriptor* rather than an author's actual age. As an ecological check, we draw 450 posts from the Blog Authorship Corpus [5] (2004 blogs with self-reported author age): 75 per non-neutral class in each of a teen (13–17) and adult (33–47) band. Because this corpus has no neutral-class analog and is labeled with the same lexicon-plus-keyword router described in Section III rather than a curated dataset label, its absolute accuracy is not comparable to Table I; only the within-corpus age contrast is meaningful.

## IV. Experimental Setup

We evaluate six models spanning three vendors at two cost tiers each: OpenAI (a low-cost and a mid-tier model), Google (a low-cost and a mid-tier Gemini model), and Anthropic (Claude Haiku 4.5 and Claude Sonnet 5). All models are queried through a single OpenAI-compatible interface with a fixed zero-shot prompt (v1) that defines the four classes and requests a JSON response with a label and a one-sentence explanation. We report accuracy, macro-F1, and per-class F1 on the 300-item evaluation set; an *explanation-consistency* check on whether a model's stated rationale for a `social_anxiety` prediction actually names a social cue; and, on the 272-item template set, accuracy and prediction-flip rate broken out by demographic group and register. We additionally run a second prompt (v2: school-counselor role framing plus one few-shot example per class) on the same 300 items, and run the Blog Corpus set under v1. As a supervised reference, we fine-tune DistilBERT [6] (66M parameters, three epochs) on 300 examples per class from the same sources, disjoint from the evaluation set. To make sample size explicit, we report 95%

TABLE I
OVERALL ACCURACY, MACRO-F1, AND SOCIAL_ANXIETY RECALL (300-ITEM BALANCED SET). 95% WILSON INTERVALS ARE ABOUT $\pm 0.05$ FOR ACCURACY ($n = 300$) AND $\pm 0.11$ FOR SA RECALL ($n = 75$; $\pm 0.07$ AT 0.89). DISTILBERT ROW IS A FINE-TUNED SUPERVISED BASELINE, NOT ZERO-SHOT.

| Model | Accuracy | Macro F1 | SA Recall |
|---|---|---|---|
| GPT (low tier) | 0.63 | 0.62 | 0.38 |
| GPT (mid tier) | 0.71 | 0.71 | 0.45 |
| Gemini (low tier) | 0.70 | 0.70 | 0.60 |
| Gemini (mid tier) | 0.65 | 0.65 | 0.44 |
| Claude Haiku 4.5 | 0.66 | 0.66 | 0.47 |
| Claude Sonnet 5 | 0.70 | 0.69 | 0.47 |
| DistilBERT (fine-tuned) | 0.79 | 0.79 | 0.89 |

Wilson score intervals for proportions (accuracy, recall, flip rate) computed from each metric's $n$.

## V. RESULTS

### A. Classification performance

Table I summarizes the main result. Sadness is the easiest class for every model (F1 0.75–0.86); social anxiety is the hardest or near-hardest for every model, with **recall between 0.38 and 0.60** – meaning 40–62% of true social-anxiety examples are missed. Confusion matrices show the dominant error mode is not confusing social anxiety with general anxiety or sadness specifically, but predicting `neutral`: across models, 15–32% of true social-anxiety items are predicted neutral, consistent with a "default to a safe answer under uncertainty" pattern rather than confident misclassification into an adjacent emotional category. This error mode also argues against the neutral class's genre difference (Section VII) driving the result: a model keying on genre would separate EmpatheticDialogues situations *from* neutral Reddit comments, not assign them to it.

The gap is robust to sampling error: even the best zero-shot social-anxiety recall (0.60, 95% CI 0.49–0.70) lies below the DistilBERT interval (0.80–0.94).

**Moving to a more capable tier does not reliably close the gap.** GPT improves from its low to mid tier (accuracy 0.63 → 0.71), but Gemini's mid-tier model does not improve (0.70 → 0.65), and Claude's social-anxiety F1 is identical across tiers (0.57 both, despite gains on the other three classes). These tier differences are of the same order as the $\pm 0.05$ accuracy interval, so we read them as "no consistent tier benefit" rather than as a ranking of models.

**Explanation consistency.** When a model predicts social anxiety, its stated explanation names an explicit social cue (e.g., peer judgment, embarrassment) in 97.5–100% of cases across all six models, indicating that surface-level reasoning, when correct, is not fabricated post hoc.

**A fine-tuned baseline closes the gap that zero-shot prompting cannot.** DistilBERT reaches 0.79 accuracy and **0.89 social-anxiety recall** (Table I), exceeding every zero-shot LLM on the class where they struggle most. It is trained and evaluated on the same imperfect labels (Section VII), but shows the signal in this data *is* learnable from a few hundred examples, so the LLM gap is better attributed to zero-shot prompting than to label noise alone.

TABLE II
AMBIGUOUS-TEMPLATE SUBSET: DEMOGRAPHIC FLIP RATE AND REGISTER ACCURACY GAP.

| Model | Flip rate | Formal − teen-register acc. |
|---|---|---|
| GPT (low tier) | 20% | +1.4 pts |
| GPT (mid tier) | 5% | +7.5 pts |
| Gemini (low tier) | 5% | 0 pts |
| Gemini (mid tier) | 10% | 0 pts |
| Claude Haiku 4.5 | 5% | −2.5 pts |
| Claude Sonnet 5 | 5% | +7.5 pts |

TABLE III
AMBIGUOUS-TEMPLATE ACCURACY BY AGE×GENDER CELL ($n = 17$–$20$/CELL).

| Model | Adult F | Adult M | Teen F | Teen M |
|---|---|---|---|---|
| GPT (low tier) | 0.647 | 0.667 | 0.611 | 0.579 |
| GPT (mid tier) | 0.850 | 0.850 | 0.900 | 0.850 |
| Gemini (low tier) | 0.600 | 0.600 | 0.600 | 0.600 |
| Gemini (mid tier) | 0.600 | 0.600 | 0.600 | 0.600 |
| Claude Haiku 4.5 | 0.700 | 0.650 | 0.700 | 0.700 |
| Claude Sonnet 5 | 0.650 | 0.650 | 0.650 | 0.700 |

### B. Demographic and register bias probe

On the 192 unambiguous templates, all six models scored 100% accuracy and 0% flip rate across all 48 scenario-register units. On the 80 ambiguous stimuli (Section III-B), accuracy dropped to 60–87.5% and the flip rate rose to **5–20%** (Table II), i.e., 1 to 4 of 20 units per model. This confirms the initial null result was a ceiling effect rather than evidence of invariance. With only 20 units per model, however, the intervals are wide (1/20: 95% CI 1–24%; 4/20: 8–42%), so the probe can detect *that* predictions are unstable but cannot reliably rank models by flip rate.

A non-zero flip rate is consistent with either systematic bias or simple inconsistency. Table III breaks accuracy out by age×gender cell to distinguish the two, and shows no direction shared by all six models: the low-tier GPT model is the only one where both adult cells outscore both teen cells (by up to 8.8 points), while other models show a different cell as the outlier or no difference at all (both Gemini models score identically in all four cells). We do not find evidence of a consistent, cross-vendor age or gender bias. The most consistent pattern is a **register effect**: two of six models (GPT mid-tier, Claude Sonnet 5) score 7.5 points lower on teen-digital-register phrasing than on formal-register phrasing of identical content, while the two Gemini models show no gap and Claude Haiku shows a small reversed gap. A 7.5-point gap is 3 of 40 items, well inside the $\pm 14$-point interval for a single register's accuracy at $n = 40$; the probe is underpowered for effects this size, and we report it as a lead, not a finding. Its recurrence across two vendors and its relevance to real teen text motivate a follow-up with several

TABLE IV
ZERO-SHOT (V1) VS. ROLE+FEW-SHOT (V2) PROMPT ON THE SAME 300 ITEMS.

| Model | v1 Acc | v2 Acc | v1 SA-R | v2 SA-R | Agree |
|---|---|---|---|---|---|
| GPT (low) | 0.63 | 0.58 | 0.38 | 0.27 | 77.3% |
| GPT (mid) | 0.71 | 0.64 | 0.45 | 0.37 | 86.3% |
| Gemini (low) | 0.70 | 0.63 | 0.60 | 0.49 | 84.7% |
| Gemini (mid) | 0.65 | 0.61 | 0.44 | 0.40 | 86.3% |
| Claude Haiku | 0.66 | 0.60 | 0.47 | 0.36 | 79.7% |
| Claude Sonnet | 0.70 | 0.64 | 0.47 | 0.37 | 87.3% |

TABLE V
BLOG CORPUS ACCURACY BY REAL AUTHOR AGE BAND ($n = 225$ EACH).

| Model | Acc. (13–17) | Acc. (33–47) | Gap |
|---|---|---|---|
| GPT (low) | 0.349 | 0.255 | +9.4 |
| GPT (mid) | 0.448 | 0.317 | +13.1 |
| Gemini (low) | 0.391 | 0.222 | +16.9 |
| Gemini (mid) | 0.391 | 0.236 | +15.5 |
| Claude Haiku | 0.382 | 0.244 | +13.8 |
| Claude Sonnet | 0.406 | 0.240 | +16.6 |

hundred ambiguous stimuli and a paired test (e.g., McNemar) on matched formal/teen items.

### C. Prompt sensitivity: v1 vs. v2

LLM text-classification performance is known to be sensitive to prompt formulation [10]; Table IV compares the zero-shot prompt (v1, used above) with a role-framed, few-shot prompt (v2) on the same 300 items. Concretely, v2 opens with a persona framing – *"You are a school counselor trained to notice signs of social anxiety in how teens describe their day. You are NOT diagnosing anyone – this is a research text-classification task, not a clinical assessment"* – followed by one worked example per class (e.g., *"She skipped the lunch table again because she didn't know where to sit without everyone staring"* → `social_anxiety`) before the item to classify; v1 gives only the four class definitions, with no persona and no examples. v1–v2 agreement ranges from 77.3% to 87.3%, confirming predictions are prompt-sensitive. Counter to the usual expectation that role framing and worked examples help, **v2 is worse than v1 for every one of the six models, on both overall accuracy and social-anxiety recall**: accuracy drops by 0.04–0.07 and social-anxiety recall drops by 0.04–0.11. The few-shot examples and counselor framing appear to anchor predictions toward the four example patterns rather than generalizing, and may shift the model toward a more conservative (i.e., more `neutral`-biased) decision rule under an explicitly clinical-adjacent framing – itself a relevant finding for anyone tempted to add a "be a counselor" system prompt to a teen-facing classifier expecting it to help.

### D. Ecological check: real teen vs. adult authors

Table V reports accuracy on the Blog Corpus set (Section III-C), split by the author's real, self-reported age band. Absolute accuracy is low across the board (0.222–0.448) compared to the main evaluation set (0.63–0.71, Table I), consistent with the cruder, lexicon-only labels used here; the informative comparison is the within-corpus gap between age bands, not the absolute level. On that comparison, **every model is more accurate on real teen-authored text than on real adult-authored text**, by 9.4 to 16.9 points (e.g., GPT low-tier: 0.349 vs. 0.255), with balanced classes in both age bands (75 items per class per band) ruling out a trivial class-imbalance explanation.

This is the opposite direction from the register effect in Section V-B, where teen-*digital-register phrasing of identical content* was harder, not easier, for two of six models. The two results are not directly contradictory: one holds content fixed and varies register, the other holds age fixed to whatever a real author wrote. But the discrepancy is worth flagging rather than glossing over. Three explanations seem plausible and are not mutually exclusive: real teen blog text may differ from our teen-digital-register templates in more than register (topic, length, explicitness); the cruder lexicon-based labels used here may interact with age-correlated writing style in a way that has nothing to do with genuine model confusion; and the corpus itself predates smartphone-era teen writing (posts are from 2004), so its "teen" register may not resemble how a teenager writes today. We treat the age gap as a lead for follow-up, not a resolved finding.

## VI. DISCUSSION

Social-anxiety recognition is a specific, persistent weak point across every vendor and tier tested, not a generic capability gap. Models correctly pick up on a general negative emotional tone but default to `neutral` rather than resolving why the speaker is anxious. Distinguishing fear of social evaluation from a concrete non-social threat requires an implicit relational judgment that is not lexically marked the way sadness or fear typically is, which fits our finding that sadness is uniformly easy while social anxiety is not.

The bias probe results carry a methodological point beyond this task: a counterfactual test built only from clear, unambiguous examples can produce a false sense of fairness, because difficulty and bias-sensitivity are not independent. A well-designed probe needs a difficulty gradient, not only clean minimal pairs. Once we added one, a register effect – not age or gender – turned out to be the most consistent signal, and it is arguably the more actionable one for teen-facing systems: if confirmed at scale, models would be less reliable on authentic teen digital vernacular than on formal English, which matters directly for any moderation or wellness tool reading real teen-written text rather than adult-paraphrased summaries of it.

**Implications for teen-facing systems.** Four practical points follow. First, a zero-shot LLM used as a social-anxiety detector would miss roughly half of true cases, and because misses default to `neutral`, they fail *silently* rather than as visible errors; such a component should flag text for a human, never decide on its own. Second, prompt choices that look helpful (a counselor persona, worked examples) must be measured, not assumed: here they hurt. Third, a small supervised model is

cheaper and stronger on this signal than any zero-shot LLM, provided its training labels are audited. Fourth, the label set is closed. Every input is forced into one of four classes, so out-of-scope text – crisis or self-harm language, other conditions, or off-topic messages – will be silently mapped to the nearest class, usually `neutral` or `sadness`. A deployed system needs an explicit *other/abstain* option and a separate crisis-detection layer that runs before, and independently of, this classifier.

## VII. Limitations

All labels are proxy constructs derived from crowd-written or social-media text, not clinical ground truth. We ran a human audit: two raters (the author and a research mentor) independently judged the same 40-item sample (10/class) from the main evaluation set. Agreement with the assigned label was 80.0% and 87.5% respectively, and the two raters agreed with *each other* on only 70% of items – itself evidence that these proxy labels are not a clean, objective gold standard. Disagreement concentrated on `social_anxiety` for both raters (75% and 60% agreement, their lowest of any class): e.g., *"Big day next week and I can't wait. My daughter will be in an Honors class her freshman year"* is labeled `social_anxiety` (routed from an ED self-report of "anxious") but reads as proud or excited, not anxious – label noise already present in the source self-report before our routing rule ever saw it. The `neutral` class is drawn from a different corpus (GoEmotions) than the other three classes (EmpatheticDialogues), so it differs from them in genre, not only emotional content: 20.0 words with a first-person pronoun 96.9% of the time, versus 12.6 words and 41.3% (Section III) – a classifier could partly learn this length/register fingerprint rather than pure emotional content; a neutral control drawn from the same source (e.g., low-intensity EmpatheticDialogues situations) would separate the two. The ambiguous-template results rest on a small sample ($n = 80$). The Blog Corpus comparison complements, not confirms, the synthetic probe (caveats in Section V-D). We tested only two prompts; a larger sweep could clarify whether the v2 degradation (Section V-C) generalizes. The DistilBERT baseline trains and evaluates on the same proxy labels as everything else, so its recall shows the signal is learnable, not that the labels are correct; it may partly learn the social-keyword routing rule itself. **Transfer.** All data are English and drawn from US-centric sources (crowdworker dialogues, Reddit, U.S. blogs). How social anxiety is expressed varies across cultures (e.g., *taijin kyofusho*, which centers on fear of offending others), languages, and platforms, and the drop from the main set to blog text (Section V-D) already shows sensitivity to domain shift. We make no claim that these results transfer to other languages, cultures, or chat platforms.

## VIII. Conclusion and Future Work

Across six LLMs from three vendors, social-anxiety recall lags well behind sadness and does not improve reliably with model tier, though explanations, when correct, cite genuine social cues. A bias probe built from clean examples found no demographic effect – an artifact of difficulty, not fairness – while ambiguous scenarios revealed 5–20% prediction instability, no consistent age or gender direction, and a tentative register effect. A role-framed few-shot prompt hurt every model, and a small fine-tuned model closed most of the gap. Future work should scale the evaluation and ambiguous-template sets with paired significance tests; add a same-source neutral control and a domain-trained rater to the label audit; use modern teen-authored text in place of the 2004 Blog Corpus; and extend to non-English and non-U.S. text.

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